

ATKIN, MN, USA

WMO: 727504

Lat: 46.548N

Lon: 93.677W

Elev: 367

StdP: 96.99

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 94999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-28.9	-26.0	-33.3	0.2	-28.3	-29.9	0.2	-24.2	8.3	-8.0	7.5	-8.0	0.7	250	0.569

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.6	29.9	23.5	28.0	22.0	27.2	21.1	24.9	28.5	23.6	27.2	22.2	25.9	4.0	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.7	19.4	27.6	22.4	17.9	26.5	21.1	16.5	25.1	77.7	28.6	72.5	26.9	66.9	25.1	28.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.0	7.2	6.2	DB	-33.3	32.5	3.9	1.4	-36.2	33.6	-38.5	34.4	-40.7	35.2	-43.5	36.2	(4)
(5)			WB	-33.4	26.7	3.8	1.4	-36.2	27.7	-38.4	28.5	-40.5	29.3	-43.3	30.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.0	-12.0	-9.6	-2.4	4.9	11.6	16.9	19.7	18.1	14.0	7.0	-1.0	-8.3		
	DBStd	12.25	7.89	7.34	7.12	5.21	4.50	3.83	3.38	3.27	4.77	4.91	5.93	6.79		
	HDD10.0	2853	682	549	386	166	33	2	0	0	16	121	333	567		
	HDD18.3	5022	940	782	642	402	214	69	24	44	144	352	581	826		
	CDD10.0	1023	0	0	2	13	83	210	299	250	136	27	2	0		
	CDD18.3	151	0	0	0	1	6	27	66	36	15	1	0	0		
	CDH23.3	1662	0	0	0	6	108	316	699	393	131	8	0	0		
	CDH26.7	371	0	0	0	1	24	64	180	79	22	1	0	0		
(14) Wind	WSAvg	2.5	2.4	2.6	2.7	3.0	2.9	2.4	2.1	2.0	2.4	2.7	2.7	2.5		
(15) Precipitation	PrecAvg	728	20	17	33	59	79	111	111	84	77	78	37	22		
	PrecMax	839	53	42	82	165	156	188	203	161	151	169	97	42		
	PrecMin	499	4	3	14	16	34	37	48	36	22	18	3	8		
	PrecStd	90	13	9	17	31	31	37	39	37	33	40	26	10		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.3	8.8	18.7	23.8	29.2	30.9	32.4	31.3	28.9	24.1	17.5	7.5		
		MCWB	3.2	5.8	12.6	14.3	19.7	23.4	25.6	24.7	23.3	16.4	11.9	4.7		
	2%	DB	2.7	5.7	13.8	20.9	26.4	28.1	30.2	28.7	27.1	21.1	13.0	4.1		
		MCWB	1.1	3.0	8.5	12.3	17.3	21.6	24.3	23.0	21.2	15.8	8.8	2.1		
	5%	DB	1.1	2.8	10.2	17.6	23.7	27.1	28.6	27.3	24.1	17.7	10.1	2.5		
		MCWB	-0.1	1.1	6.7	10.1	15.5	20.3	23.5	21.3	19.3	12.9	7.1	1.2		
	10%	DB	-1.1	1.2	7.3	15.0	21.2	25.0	27.3	26.0	22.5	15.4	7.6	1.1		
		MCWB	-1.8	-0.1	4.4	8.6	14.1	18.8	21.9	20.3	18.2	11.5	5.3	0.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.8	5.6	14.0	16.1	22.9	25.1	27.3	26.0	24.1	19.3	13.0	4.9		
		MCDB	6.2	8.5	17.0	21.1	26.5	29.0	30.4	29.4	27.9	21.5	15.8	7.0		
	2%	WB	1.3	3.4	9.7	12.9	19.1	23.4	25.6	24.0	22.1	15.9	9.4	2.7		
		MCDB	2.7	5.1	13.2	18.5	23.8	27.0	29.1	27.7	25.6	19.6	13.0	4.1		
	5%	WB	-0.2	1.5	7.0	11.1	16.9	21.6	24.1	22.3	20.5	13.9	7.4	1.0		
		MCDB	0.8	2.9	9.9	16.9	21.5	25.4	27.6	26.0	23.7	17.8	9.5	2.3		
	10%	WB	-2.0	-0.2	4.4	9.0	15.2	20.0	22.7	21.0	19.0	11.9	5.3	-0.1		
		MCDB	-1.4	1.1	7.3	14.4	19.9	23.8	26.3	24.5	21.8	14.5	7.8	0.9		
(35) Mean Daily Temperature Range	MDBR		10.5	11.4	11.4	12.4	13.5	12.7	12.6	12.9	12.5	10.6	8.9	8.7		
	5% DB	MCDBR	10.1	11.3	14.3	17.8	17.8	14.7	13.6	14.3	14.2	14.9	13.3	9.7		
		MCWBR	8.7	9.0	9.6	10.6	10.1	9.0	8.8	9.3	9.5	9.5	9.4	8.1		
	5% WB	MCDBR	9.4	10.2	13.5	15.9	14.6	12.6	12.6	12.8	12.4	12.8	12.1	9.2		
		MCWBR	8.3	8.2	9.6	10.3	9.7	8.8	9.1	9.3	9.0	9.0	9.1	7.8		
(40) Clear-Sky Solar Irradiance	taub		0.251	0.270	0.307	0.361	0.382	0.377	0.379	0.387	0.358	0.310	0.285	0.260		
	taud		2.385	2.340	2.380	2.336	2.331	2.372	2.390	2.382	2.438	2.567	2.512	2.408		
	Ebn at Noon		864	912	922	895	884	888	881	859	855	860	818	807		
	Edh at Noon		69	90	101	117	122	118	114	111	96	73	62	61		
(44) All-Sky Solar Radiation	RadAvg		1.58	2.70	3.81	4.68	5.34	5.81	6.19	5.18	3.90	2.39	1.48	1.11		
	RadStd		0.17	0.24	0.37	0.44	0.40	0.39	0.25	0.41	0.29	0.33	0.15	0.11		

Historical Trends

		DBAvg		Heating		Cooling		Degree-Days			
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (54 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page